

Automatic Air Cooling System using DHT11 Temperature and Humidity Sensor

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ABSTRACT

This study presents the development and simulation of an Automatic Air Cooling System using a DHT11 Temperature and Humidity Sensor to implement sensor-based environmental control that automatically activates cooling when heat index levels exceed predefined thresholds. The system integrates a DHT11 sensor with threshold-based decision logic, a 128×64 LCD display, a DC cooling fan, and an Arduino Nano microcontroller to continuously monitor temperature and humidity in real time and regulate the cooling fan without manual intervention.

A developmental research design was employed, and the complete system was modeled and tested in the Tinkercad simulation environment. A total of 10 structured test cases were conducted, evaluating sensor responsiveness, heat index computation accuracy, fan activation performance, and LCD display feedback. Results showed consistent and reliable activation of the cooling fan at critical heat index levels, accurate real-time visualization of environmental conditions, and continuous autonomous system operation.

The study's key contribution is demonstrating a simulation-driven approach for developing low-cost and automated cooling systems that intelligently integrate temperature and humidity sensing. Future enhancements include adding wireless connectivity, IoT monitoring, multiple cooling mechanisms, data logging, and remote control capabilities.

Keywords: *Automatic Air Cooling System, DHT11 Sensor, Heat Index Monitoring, Arduino Nano, 128×64 LCD, Temperature and Humidity Control, Embedded Systems, Automation*

INTRODUCTION

High-temperature and high-humidity environments can cause thermal discomfort, reduce productivity, and stress equipment. In tropical countries like the Philippines, prolonged exposure to high heat index levels is common in households, classrooms, and workplaces, posing health risks and lowering overall comfort. Perceived heat is influenced not only by temperature but also by moisture in the air, a factor that conventional cooling systems often fail to consider.

Traditional systems relying solely on temperature-based control and manual operation are often inefficient—they may fail to provide cooling when needed or consume excessive energy by running unnecessarily. Many existing solutions also lack real-time humidity monitoring, automated fan activation, or fully local operation, with some dependent on internet connectivity. The current study addresses these limitations by integrating a DHT11 temperature and humidity sensor, Arduino Nano, driver circuitry, DC fan, and 128×64 LCD, enabling automatic fan activation based on heat index, real-time local monitoring, and fully autonomous operation without internet reliance.

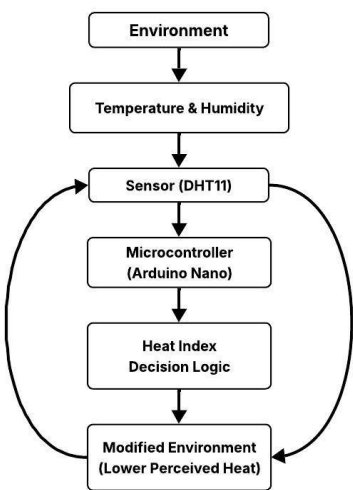
In response, this study developed an Automatic Air Cooling System that continuously monitors environmental conditions, calculates the heat index, and activates the fan when thresholds are exceeded. By employing an automated, sensor-driven approach in a low-cost simulation environment, the system provides an accessible, efficient, and practical solution for maintaining thermal comfort in hot and humid settings.

REVIEW OF RELEVANT THEORY, STUDIES, AND LITERATURE

Theoretical Framework

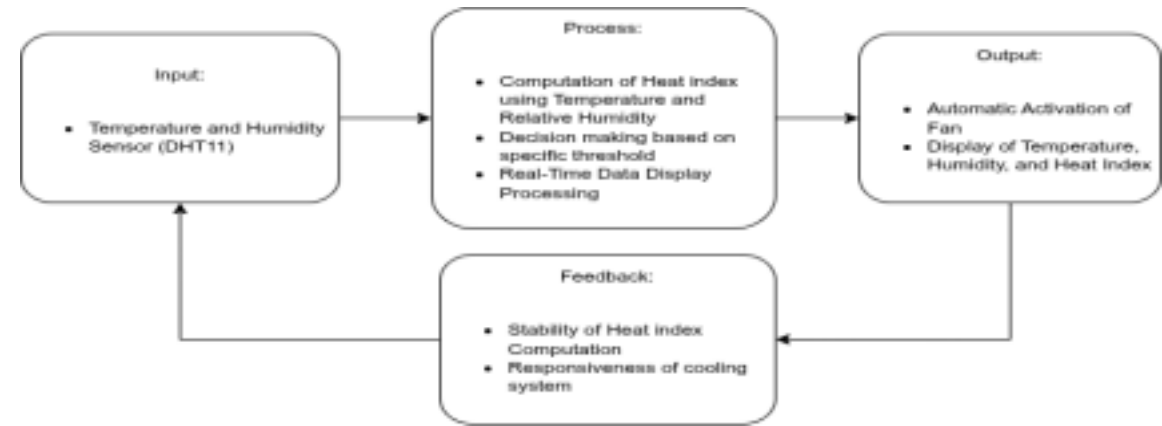
The theoretical foundation of this study is based on heat index theory, automatic control systems, and embedded system design. The heat index represents the combined effect of temperature and humidity on perceived heat, making it a more reliable indicator of thermal discomfort than temperature alone. Automatic control theory supports the use of feedback mechanisms where sensor data is continuously monitored and processed to trigger appropriate system responses without manual input.

Figure 1: System Theory



System theory applied to the Automatic Air Cooling System using a DHT11 Temperature and Humidity Sensor. The system operates as a closed-loop automated control system in which environmental conditions are continuously sensed, analyzed, and acted upon. Temperature and humidity data are collected from the environment through the DHT11 sensor and processed by the microcontroller to determine appropriate cooling actions. Based on the processed sensor data, the cooling devices are automatically activated or deactivated. Feedback from the environment enables the system to continuously adapt its operation without the need for manual intervention.

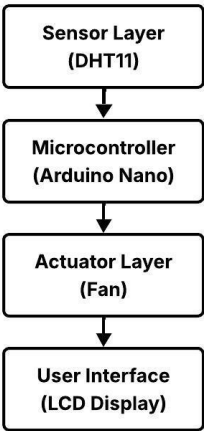
Figure 2: Input–Process–Output (IPO) Model



The Input–Process–Output (IPO) Model illustrates the functional operation of the Automatic Air Cooling System using a DHT11 Temperature and Humidity Sensor. During the input stage, the DHT11 sensor acquires real-time ambient temperature and relative humidity data from the environment. In the process stage, the Arduino Uno microcontroller collects and analyzes the sensor readings to determine the appropriate system response based on predefined threshold values. The control logic evaluates whether the measured temperature and humidity levels fall within safe or critical ranges to trigger cooling actions.

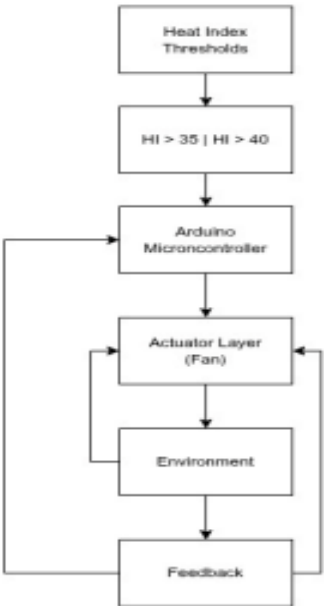
In the output stage, the system automatically activates the cooling fan to improve thermal comfort when critical environmental conditions are detected. The LCD module simultaneously provides real-time visualization of temperature and humidity values for user awareness. This IPO model ensures responsive and energy-efficient system operation by continuously monitoring environmental conditions and automatically adjusting cooling actions without manual intervention

Figure 3: Embedded Systems Theory



In the embedded system architecture of the Automatic Air Cooling System using a DHT11 Temperature and Humidity Sensor, the integration of hardware and software components is highlighted. The Arduino Uno functions as the embedded controller, responsible for real-time data processing, decision-making, and actuator control based on inputs from the temperature and humidity sensor.

Figure 4: Control Systems Theory

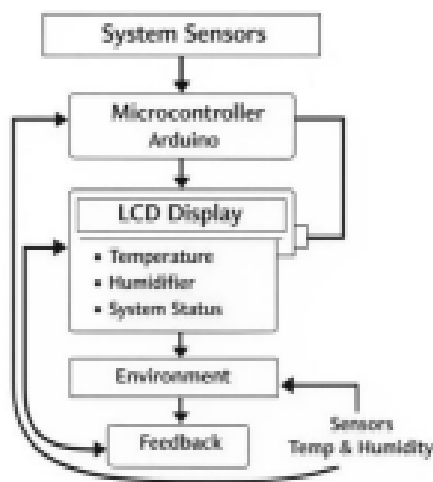


In the control system of the Automatic Air-Cooling System using a DHT11 Temperature and Humidity Sensor, the system operates as a closed-loop feedback control system. Temperature and humidity data collected from the environment are processed by the Arduino microcontroller, which evaluates the readings against predefined threshold ranges to determine appropriate cooling actions.

When the measured temperature and humidity indicate low cooling demand, the controller activates the fan minimally. For moderate environmental conditions, the LCD display issues a warning, and when critical levels are reached, the system increases fan operation to enhance cooling and reduce thermal discomfort.

The resulting changes in environmental conditions are continuously sensed and fed back into the system, enabling automatic and adaptive regulation without manual intervention. This feedback-based control strategy ensures efficient cooling performance while minimizing unnecessary energy consumption.

Figure 5: Human Computer Interaction (HCI)



Human-Computer Interaction of the Automatic Air-Cooling System using a DHT11 Temperature and Humidity Sensor. Although the system operates automatically, users can monitor real-time temperature and humidity readings, fan status, and system performance through the LCD interface. This provides transparency, facilitates easy monitoring, and allows users to remain informed without requiring manual control.

Framework Summary

The system framework follows an input-process-output model. Temperature and humidity data serve as inputs, which are processed by the microcontroller to calculate the heat index. When the computed value exceeds a set threshold, output devices (fan) are activated automatically. This closed-loop framework ensures timely response, efficient cooling, and reduced energy consumption.

RELATED LITERATURE

Temperature-Based Fan Control Systems

Research by Kumar et al. (2022) demonstrates the effectiveness of using temperature sensors, such as the LM35, to achieve automated fan control. Their findings highlight improved energy efficiency and environmental regulation, supporting this study's use of the DHT11 to enable reliable temperature measurement for air-cooling automation.

IoT-Enabled Environmental Monitoring

Lee and Park (2023) emphasize the benefits of integrating temperature and humidity sensors with IoT

platforms for remote monitoring and control. Their study uses DHT22 sensors to provide real-time feedback via mobile applications, which supports the concept of dual-sensor monitoring in this project while highlighting the advantage of local automation without reliance on internet connectivity.

Arduino-Based Automation Systems

Ahmed et al. (2021) introduced Arduino-based automated systems for energy-efficient cooling applications, underscoring the accessibility, reliability, and versatility of microcontrollers for real-time control. Their work validates the selection of Arduino as the primary controller for this study and aligns with the embedded system logic implemented in the current project.

Humidity-Responsive Cooling Mechanisms

Singh et al. (2020) explored air-cooling systems that respond to both temperature and humidity using DHT11 sensors. While their system provided real-time environmental monitoring, it lacked automated fan activation, requiring manual user intervention. This supports the current researchers aim to implement fully automatic cooling based on dual-sensor input.

User Interaction and Feedback in Cooling Systems

Modern cooling automation systems integrate user input controls and visual feedback mechanisms such as LEDs or displays. Studies in smart home applications show that combining user interaction with automatic sensing enhances usability, safety, and system responsiveness. This supports the design of the current study, which integrates real-time DHT11 sensing with automated fan control and optional user alerts.

Table 1. Comparison Matrix of Related Studies and Current Research

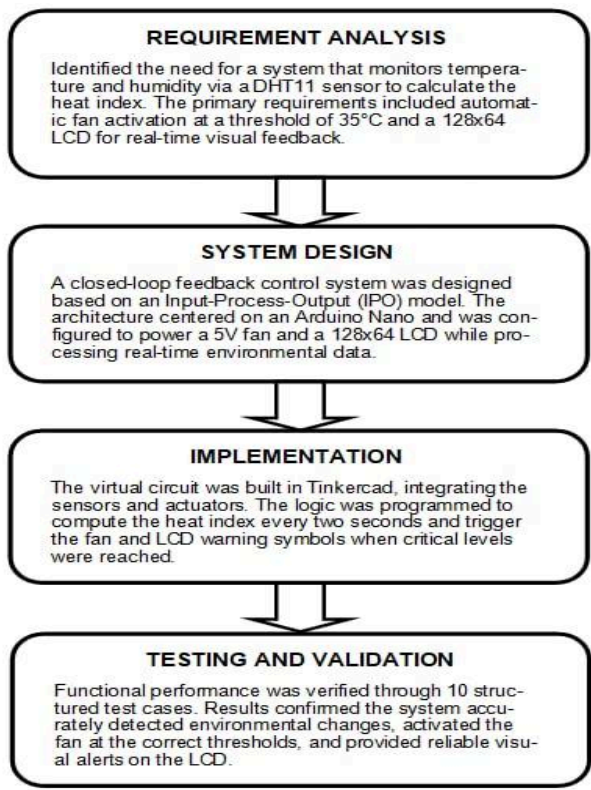
Study	Sensor Used	Platform / Technology	Key Feature(s)	Gap Addressed by This Study
Kumar et al. (2022)	LM35 Temperature Sensor	Arduino-based system	Automated fan control based on temperature	Does not account for humidity levels; limited environmental responsiveness
Lee & Park (2023)	DHT22 (Temp & Humidity)	IoT-based microcontroller	Remote monitoring and fan control via mobile app	Relies on internet connectivity; lacks full local automation
Singh et al. (2020)	DHT11	Raspberry Pi + Relay	Real-time temperature & humidity display	Lacks automated fan activation; requires manual intervention
Current Study (Automatic Air Cooling System using DHT11)	DHT11 (Temperature & Humidity)	Arduino Nano + Driver IC + LCD Module	Automatic fan activation based on computed heat index; real-time data display on LCD	Provides dual-sensor feedback, local automation, real-time environmental monitoring, and enhanced responsiveness

				without internet reliance
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METHODOLOGY

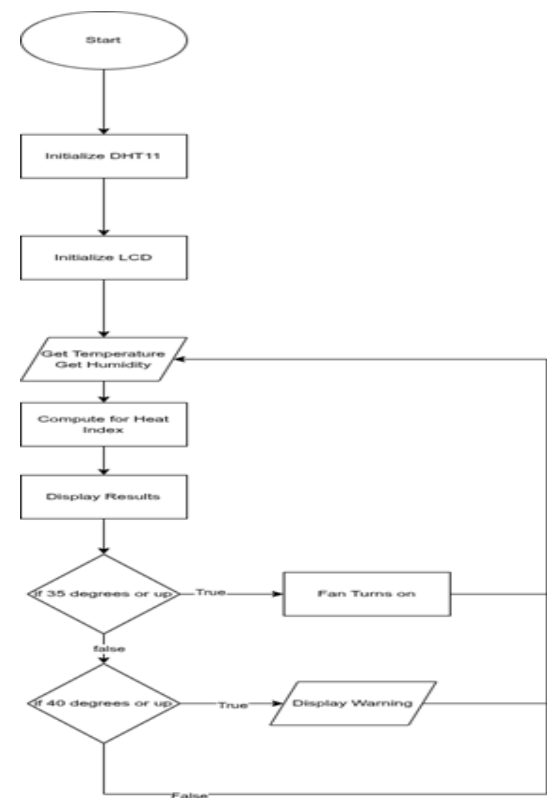
This study employed a descriptive and developmental research design focusing on system design, simulation, and functional testing. The methodology involved hardware integration, software development, and system simulation. Temperature and humidity data were collected using a DHT11 sensor and transmitted to the microcontroller every two seconds. The Arduino processed the data and calculated the heat index using programmed algorithms. When the heat index exceeded the predefined safe threshold, the system automatically activated the fan. An LCD displayed real-time environmental data and system status. System behavior was observed under different simulated environmental conditions to evaluate responsiveness and functionality.

Figure 6. Waterfall Model



The development of the Automatic Air Cooling System followed the Waterfall Model, consisting of sequential phases: Requirement Analysis, System Design, Implementation, and Testing & Validation. During the Requirement Analysis, the researchers identified both functional and non-functional requirements of the system, including automatic temperature detection, real-time data processing, efficient fan activation, user safety, energy efficiency, and system reliability. In the System Design phase, the virtual circuit was arranged in Tinkercad, and the block-code logic was outlined, incorporating sensor data acquisition, temperature value calculation, and threshold-based decision making for cooling activation. Implementation involved constructing the virtual circuit and programming the block code to read raw sensor values which are the temperature and humidity, calculate them into heat index, compare the computed values against predefined thresholds, and control the cooling fan and LCD accordingly. Finally, Testing and Validation were performed by simulating various temperature conditions in Tinkercad to ensure that the system accurately calculated temperature values, responded correctly to environmental changes, and operated safely and reliably.

Figure 7: Flow Chart

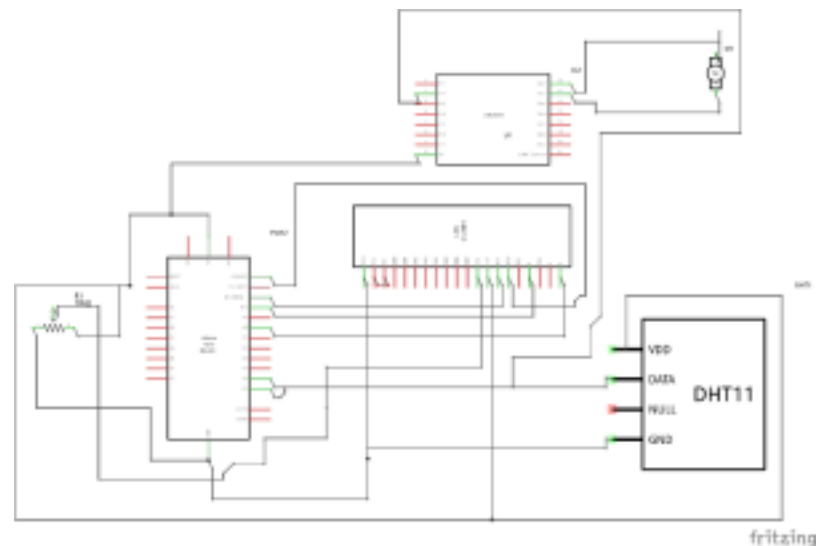


The flowchart describes the operating sequence of an Automatic Air Cooling System using a DHT11 Temperature and Humidity Sensor controlled by a microcontroller. The process begins with system initialization, during which the DHT11 sensor is configured to measure ambient temperature and humidity. These sensor readings are then processed by the microcontroller to determine appropriate cooling actions.

After initialization, the LCD display module shows the measured temperature and humidity values in real time, allowing continuous monitoring of environmental conditions. The system then evaluates the sensor readings against predefined threshold ranges. When the measured values indicate low cooling demand, the fan operates minimally. When critical levels are reached, the LCD displays a warning and the fan increases cooling. If conditions fall within normal ranges, the fan remains off and the warning is cleared.

The process continuously repeats by returning to the display and monitoring stage, enabling the system to automatically regulate cooling in response to changing environmental conditions.

Figure 7: Schematic Diagram



This schematic diagram presents the complete electrical configuration of the Automatic Air-Cooling System using a DHT11 Temperature and Humidity Sensor. The system is centered around an Arduino Nano microcontroller, which processes environmental data collected by the DHT11 sensor. The sensor measures ambient temperature and humidity, providing inputs used by the microcontroller to compute the heat index using the Adafruit DHT11 library.

Output components include a DC fan for cooling and a 128×64 LCD module for real-time display of temperature, humidity, and system status. A potentiometer is incorporated to adjust LCD contrast for optimal visibility. Fan control is managed through a driver IC, which receives logic-level signals from the Arduino and powers the DC fan via a 12V external power source, ensuring sufficient current delivery while protecting the microcontroller from electrical overload.

This configuration integrates sensing, computation, display, and actuation, enabling the system to automatically regulate cooling based on real-time environmental conditions.

RESULTS & DISCUSSION

The Automatic Air Cooling System using a DHT11 Temperature and Humidity Sensor was assessed by Tinkercad simulation and real-world testing to analyze its efficacy under various temperature, humidity, and calculated heat index circumstances. The system effectively monitored environmental parameters in real time utilizing the DHT11 sensor, computed the heat index via programmed logic, and appropriately activated the DC fan upon the heat index reaching or surpassing the critical threshold of 35 °C, while deactivating the fan when conditions remained within safe limits. The 128×64 LCD display provided prompt response by exhibiting temperature, relative humidity, heat index numbers, and warning indicators under critical conditions.

The initial experiment failed because the Arduino Uno failed to reach the power requirement for it to run, to solve this we used relays, the first experiment also used digital output which later got replaced to PWM for better fan control and also more, it will also conserve more power. In the second experiment we added the PWM and an LCD to provide display for the temperature and humidity as well as displaying and using a battery for power supply for the humidifier. The third experiment will use a module that has AMS1117 to provide power for both Arduino and the fan and humidifier the fourth experiment removes the humidifier as it requires specific power supply which this experiment cannot provide the fifth experiment happens an incident where the Arduino Uno had a failure, which prompted the researchers to find alternatives such as an Arduino Nano, which is more convenient because it's easier to be mounted on a breadboard. Lastly, table 3 illustrates that the system reliably reacted to both natural environmental changes and regulated thermal exposure, accurately turning on and off the fan in response to shifts in temperature and humidity levels. The results confirm that the system functions effectively beyond simulation, demonstrating reliable sensor-based decision making and closed-loop control in real-world conditions.

Requirements

The functional requirements of the Automatic Air Cooling System focus on its operational capabilities. The system continuously monitors ambient temperature and relative humidity using a DHT11 sensor to ensure real-time detection of environmental changes. Fan activation is allowed only when the heat index reaches or exceeds the critical threshold of 35°C, while the system remains inactive when conditions are within safe limits. The 128×64 LCD display provides immediate visual feedback, showing temperature, humidity, heat index, and warning symbols when critical levels are detected. All decision-making and control logic are implemented using the Arduino Nano and fan control, ensuring structured, sequential, and responsive system operation.

The non-functional requirements define the quality and performance aspects of the system. The Automatic Air Cooling System is designed to provide real-time responsiveness by continuously processing sensor readings

and updating fan speed and LCD display outputs immediately. Safety and efficiency are prioritized, with logic in place to prevent unnecessary fan operation under safe conditions. The LCD module offers stable and accurate feedback, reflecting the system's current state without flicker or delay. Measurement accuracy is maintained by ensuring consistent DHT11 sensor readings within the simulation range. Additionally, the fan operates reliably, activating or deactivating according to programmed heat index thresholds, guaranteeing effective and energy-efficient cooling performance.

Table 2. Variables and Conditions of the Automatic Air-Cooling System using DHT11 Temperature and Humidity Sensor

Variable / Component	Type (Input / Output)	Parameter Measured / Controlled	Condition or Range	System Response / Action
Arduino Nano	Microcontroller	Processes logic and control outputs	Operates at 5V logic; continuous loop evaluation	Reads sensor and calculates heat index to control the fan and drive the LCD display
Fan	Output	Mechanical Actuator to produce wind	Activates at 35 degrees heat index to provide cooling	Provides Cooling
DHT11	Input	Measures Temperature and Relative Humidity	Operation Range: 20% to 80% Humidity ($\pm 5\%$ accuracy), Temperature 0-50°C ($\pm 2\%$ accuracy),	Reads Temperature and sends data to arduino uno using 40 bit serial in 1Hz.
128x64 LCD	Output	Displays information	When 40°C is reached it will display a warning symbol	Displays information such as Temperature, Relative Humidity and Heat Index

Table 3. Variables and Conditions of the Automatic Air-Cooling System using DHT11 Temperature and Humidity Sensor

Test #	Input Condition	Observed Output	Expected Output	Pass/ Fail	Remarks / Behavior Explanation
1	Natural occurring Instance Heat Index = 33	No Warning Fan is off	No Warning Fan is off, the location is cool enough	Pass	System continuous the loop

	Temperature = 30.8 Humidity = 63				
2	Natural occurring Instance Heat Index = 33 Temperature = 30.8 Humidity = 71	No Warning, Fan is turned on	No Warning Fan is turned on to cool down the location	Pass	The DHT11 will detect the lowering of heat index slowly (Temperature wise)
3	Natural occurring Instance Heat index = 41 Temperature = 31.3 Humidity = 79	Warning and the fan is turned on	Warning to provide the user information that the location is at unsafe temperature	Pass	The fan continues to spin and the DHT11 continuously detects temperature.
4	Natural occurring Instance Heat index = 36.16 Temperature = 31 Humidity = 65	No Warning and the fan is turned on	The information is displayed at the LCD while the fan is turned on	Pass	The fan continues to spin and the DHT11 continuously detects temperature.
5	Soldering iron placed near DHT11 at 3cm for 5 seconds	Display warning and fan is turned on reaching Heat Index of 42 degrees while the temperature reach at 34.2 degrees and the humidity reached 50%	The warning will show in LCD and the fan will be turned on	Pass	The fan continues to spin and the DHT11 continuously detects temperature. The temperature and heat index rises, while the humidity lowers
6	Soldering iron placed near DHT11 at 3cm for 40 seconds	Displays warning, temperature reaches 42.5 while the humidity reaches 31%,	The temperature continues to rise while the while the humidity falls lower, and the heat index continues to rise	Pass	The fan continues to spin and the DHT11 continuously detects temperature. The temperature and heat index continues to rise, while the humidity continues to lower.

		and the heat index reaches 48 degrees			
7	10 seconds after Soldering iron removed	Temperature lowers to 34.6 degrees while the humidity reaches 49% and the heat index is at 39.38 degrees, the warning is removed.	The temperature lowers and the humidity rises, and the heat index lowers, the warning is removed.	Pass	The expected output is reached the temperature lowers to 34 degrees while the humidity rises back as the heat source dissipating the humid air is removed
8	1 minute after the soldering iron is removed	The temperature lowers to 32.4 degrees and the humidity is at 57% the heat index lowers to 37	The temperatures cool down further and the humidity continues to recover, while the heat index continues to lower	Pass	As expected the temperature lowers and humidity recovers, and the heat index continues to lower, although the heat index sometimes fluctuate
9	2 minutes after the soldering iron is removed	The temperature continues to lower to 31.1 degrees, while the humidity recovered, and the heat index lowers to 36 degrees	The temperature and heat index lower even further and the humidity gets higher	Pass	Within the expected output the temperature and heat index lowers and the humidity rises back up again, though the heat index still fluctuates
10	4 minutes after the soldering iron is removed	The temperatures goes back to normal temperatures at 30.20 and the humidity reaches 67%, while the heat index lowers to 34.79 degrees,	The temperature and heat index goes back to normal, while the humidity recovers slowly	Pass	The experiment variables go back to normal as expected, and the data generated by the DHT11 comes from the environment and not being affected by the soldering iron.

		which turns the fan off			
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CONCLUSIONS AND RECOMMENDATIONS

The Automatic Air Cooling System using a DHT11 Temperature and Humidity Sensor was successfully developed, simulated, and validated in the Tinkercad environment. The system consistently monitored temperature and humidity in real time, calculated the heat index accurately, and activated the cooling fan when critical thresholds were reached. The 128×64 LCD display provided immediate visual feedback on temperature, humidity, and system warnings, ensuring users were informed of environmental conditions. These results confirm that the system met all research objectives by providing reliable environmental monitoring, enabling automatic fan activation based on heat index thresholds, and delivering clear real-time feedback through the LCD. The findings highlight that low-cost, sensor-driven cooling systems can effectively enhance thermal comfort, conserve energy, and provide an accessible model for automated environmental control. To further enhance system functionality and enable real-world implementation, several improvements are recommended. Integrating additional cooling mechanisms, such as a humidifier or air conditioning, could complement the DC fan while maintaining heat index as the primary control reference. Incorporating IoT modules like ESP8266 or ESP32 would enable remote monitoring, notifications, or mobile app control. A buzzer or LED warning system could alert users to critical heat index levels, and adding data-logging capabilities could allow analysis of environmental trends over time. Future research should also consider constructing a physical prototype to validate the simulation results under real-world conditions, ensuring the system's robustness, reliability, and practical applicability.

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